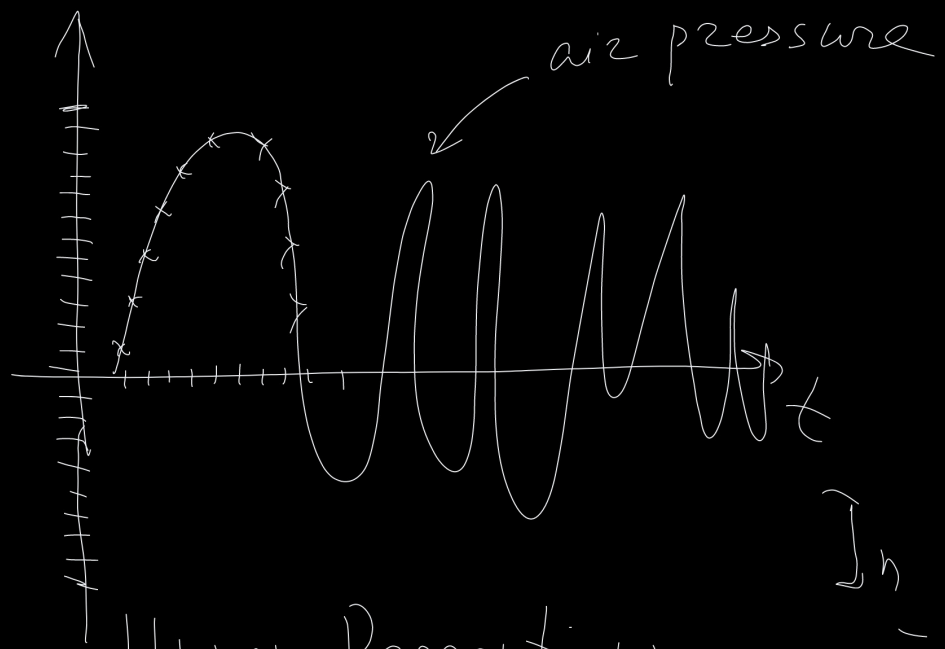
Audio Spectrograms

ASR: Audio → Spectrogram $\xrightarrow{NN}$ Text

TTS: Text $\xrightarrow{NN}$ Spectrogram $\xrightarrow{NN}$ Audio

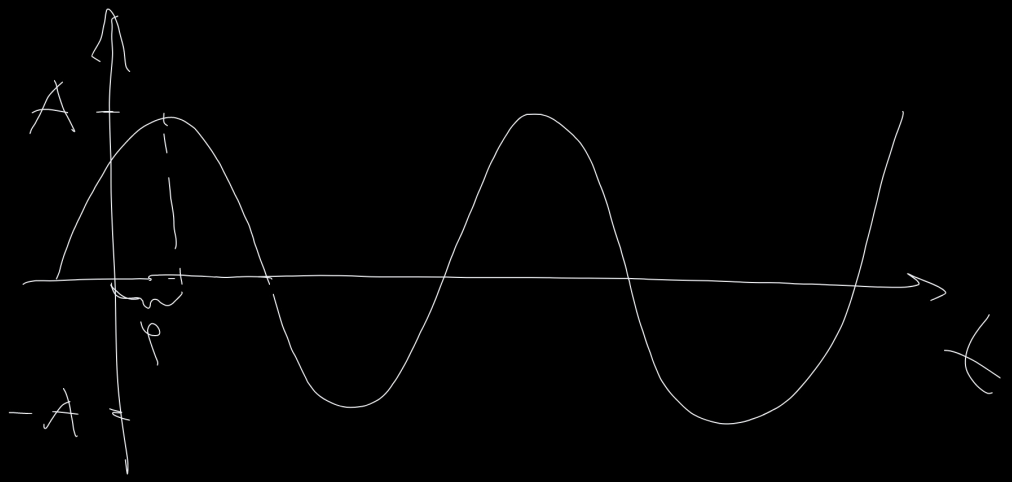


Human Perception:
20Hz – 20kHz

Audio on computer:
- sampled in time
- finite time horizon
- sampled in amplitude

In WAV file:
- sample rate, eg. 44.1 kHz
- bit depth, e.g. 8-bit, 16-bit
- # channels: mono, stereo, 5.1

$$f(t) = A\cos(\omega t + \varphi)$$



— pitch

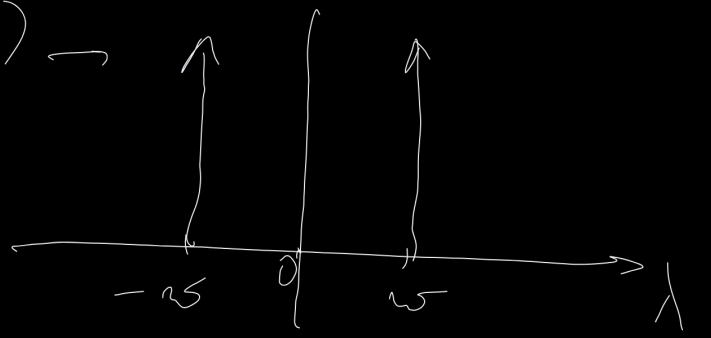
ω — frequency $\left(-\frac{\text{rad}}{\text{sec.}}\right)$

$\omega = 2\pi f$, f — Hz

A — amplitude (loudness)

φ — phase

$$f(t) = A\cos(\omega t + \varphi)$$



$$F_{\cos}(\lambda) = \int_{-\infty}^{+\infty} f(t)\cos(\lambda t)\,dt =$$

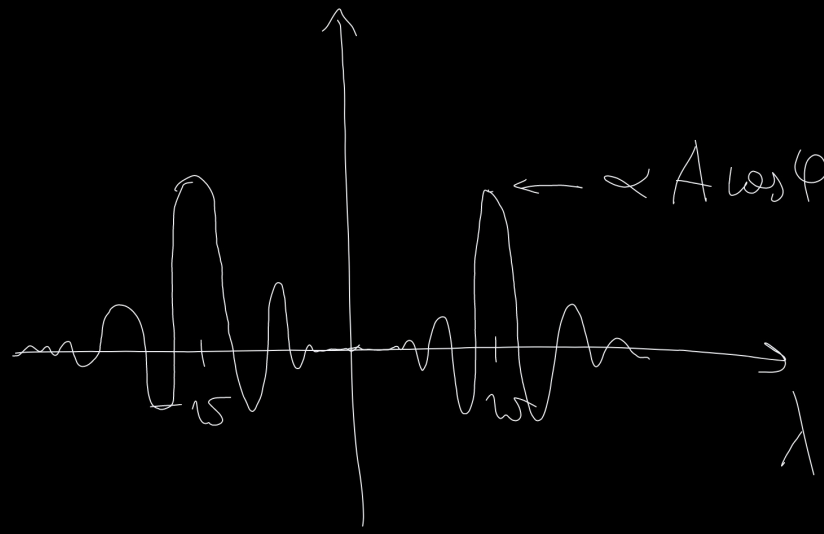
$$= A\pi\cos(\varphi)\left(\delta(\lambda-\omega)+\delta(\lambda+\omega)\right)$$

$$I(a) = \int_{-\infty}^{+\infty} \exp(-a|t|)\, f(t)\cos(\lambda t)\,dt \xrightarrow[a\to 0+]{} F_{\cos}(\lambda)$$

$a > 0$

$\vdots$

$$F_{\cos}(\lambda) = \int_{-\infty}^{+\infty} f(t)\cos(\lambda t)\,dt \approx \frac{1}{T}\int_{-T}^{T} f(t)\cos(\lambda t)\,dt \approx$$



$$\approx \frac{1}{N}\sum_{n=0}^{N-1} f\left(-T + n\frac{2T}{N}\right) \times$$

$$\times \cos\left(\lambda\left(-T + n\frac{2T}{N}\right)\right) \approx$$

$$\approx \hat{F}_{\cos}(\lambda)$$

$$F_{\cos}(\lambda) = \int f(t)\cos(\lambda t)\,dt \rightarrow A\cos(\varphi)$$

$$F_{\sin}(\lambda) = \int f(t)\sin(\lambda t)\,dt \rightarrow A\sin(\varphi)$$

$$|F(\lambda)|^2 = A^2 = F_{\cos}^2(\lambda) + F_{\sin}^2(\lambda)$$

$$\mathrm{Arg}(F(\lambda)) = \varphi = \mathrm{atan}\left(\frac{F_{\sin}(\lambda)}{F_{\cos}(\lambda)}\right)$$

$$\cos(\lambda t) - i\sin(\lambda t) = \exp(-i\lambda t)$$

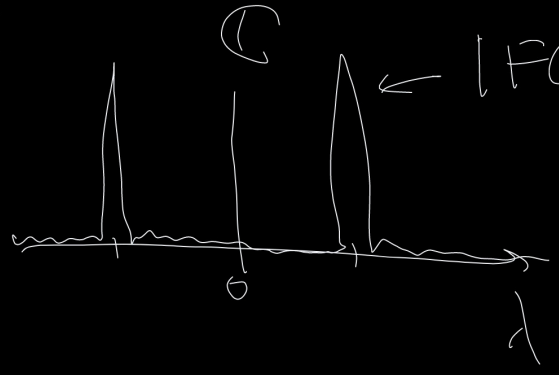
Fourier Transform: $F(\lambda) = \int_{\mathbb{C}} f(t)\exp(-i\lambda t)\,dt$

Fourier Transform

(1) inverse FT $f(t) = \frac{1}{2\pi} \int_{\mathbb{R}} F(\lambda) \exp(i\lambda t)\, d\lambda$

(2) $|F(\lambda)|^2$ – even function

$\operatorname{Arg} F(\lambda)$ – odd function



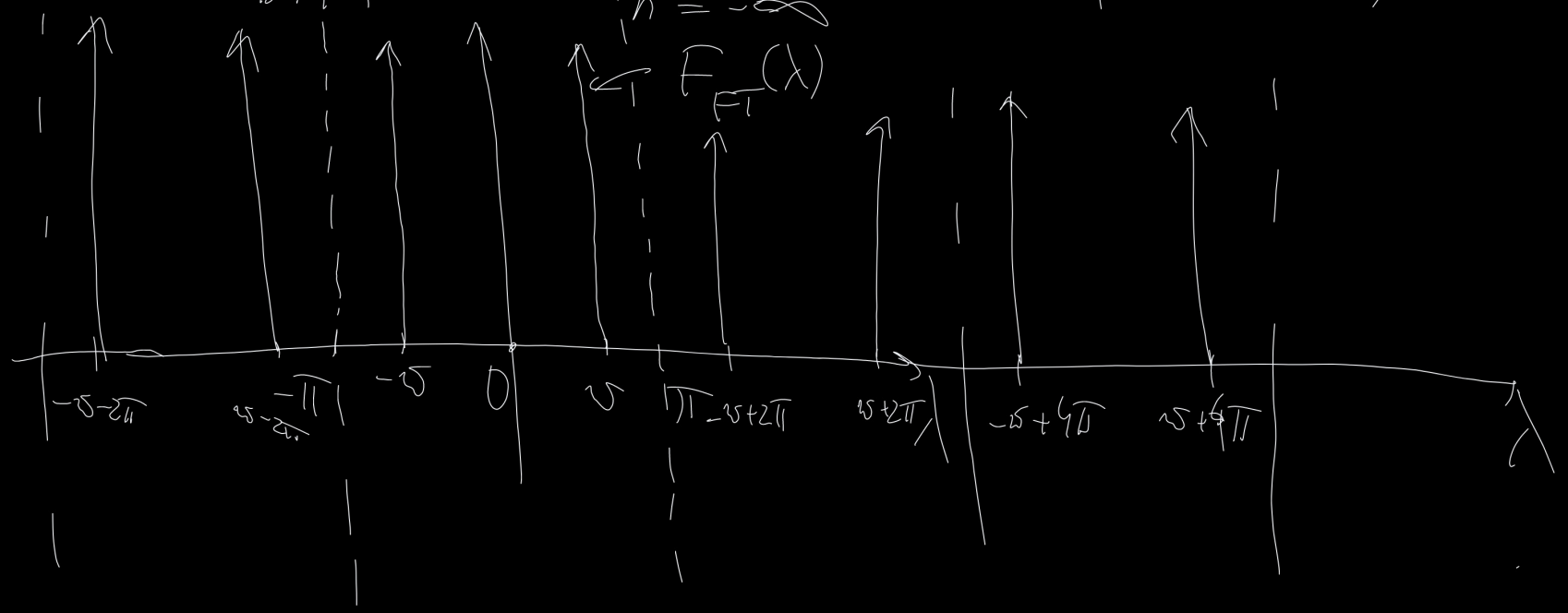
(3) $(f * g)(t) = \int_{-\infty}^{+\infty} f(t-\tau) g(\tau)\, d\tau$

$\mathcal{F}\{f * g\}(\lambda) = \mathcal{F}\{f\}(\lambda) \cdot \mathcal{F}\{g\}(\lambda)$

$\mathcal{F}\{f \cdot g\}(\lambda) = \mathcal{F}\{f\}(\lambda) * \mathcal{F}\{g\}(\lambda)$

DTFT (Discrete Time Fourier Transform)

$$F_{DTFT}(\lambda) = \sum_{n=-\infty}^{+\infty} f(n) \exp(-i \lambda n)$$



$$f_\omega(t) = \cos(\omega t)$$

$$f_\omega(n) = \cos(\omega n)$$

$$f_{\omega+2\pi k}(n) = \cos\big((\omega + 2\pi k)n\big) = \cos(\omega n)\underbrace{\cos(2\pi k n)}_{1} -$$

$$-\sin(\omega n)\underbrace{\sin(2\pi k n)}_{0} = \cos(\omega n) = f_\omega(n)$$

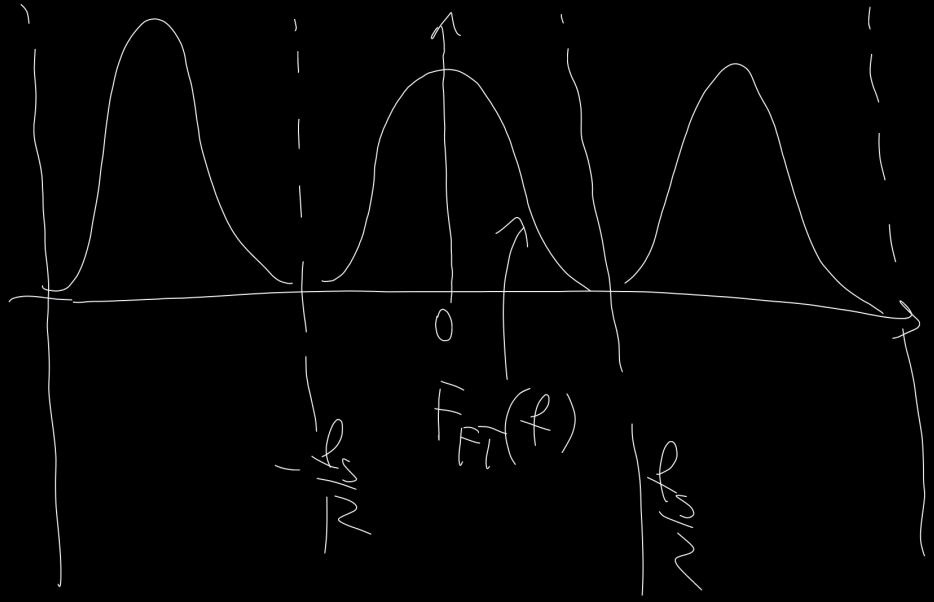
Nyquist–Shannon sampling theorem

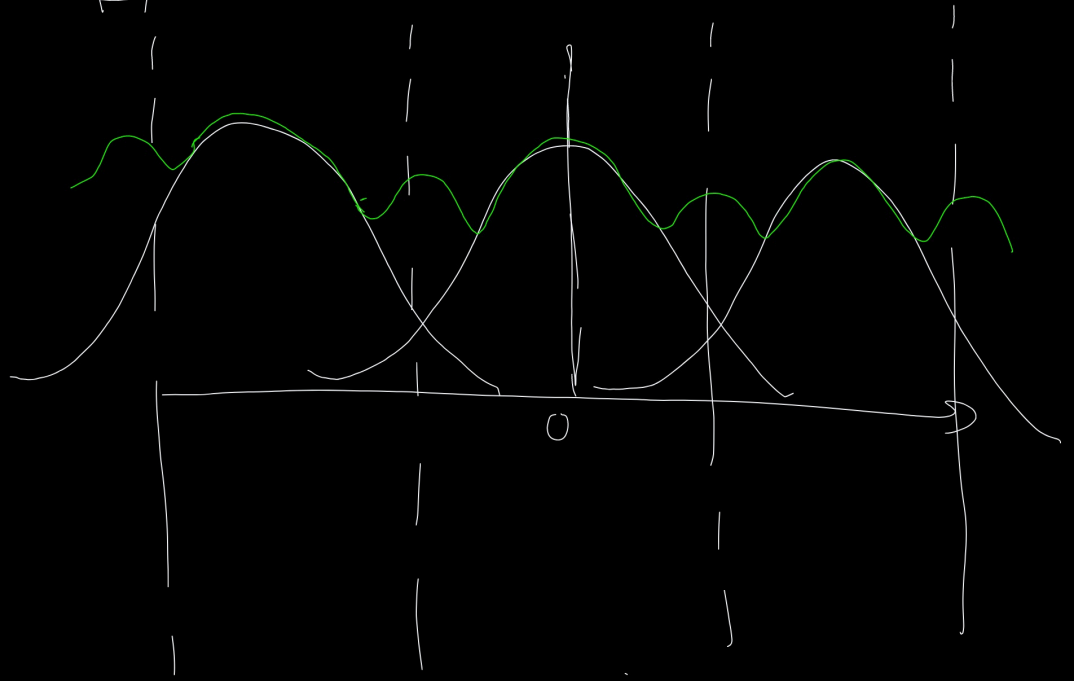
f_s – sampling freq, Hz

f_{max} – max freq, presented in $F_{FT}(\lambda)$

$f_s > 2 f_{max}$

$f_s < 2 f_{max}$





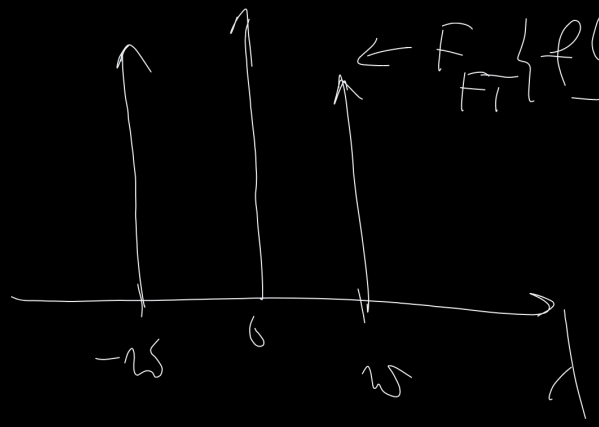
Audio resampling scheme

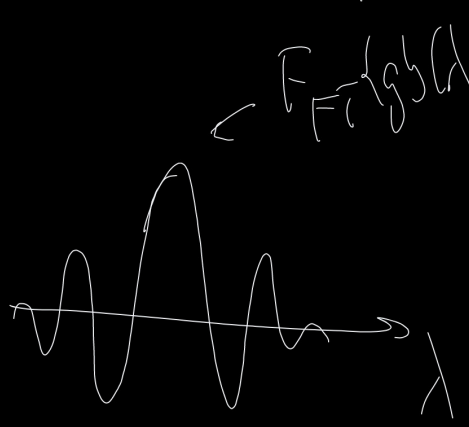
Audio in 44kHz → low-pass filter (suppress all freq. > 8kHz) → resampling → Audio in 16kHz

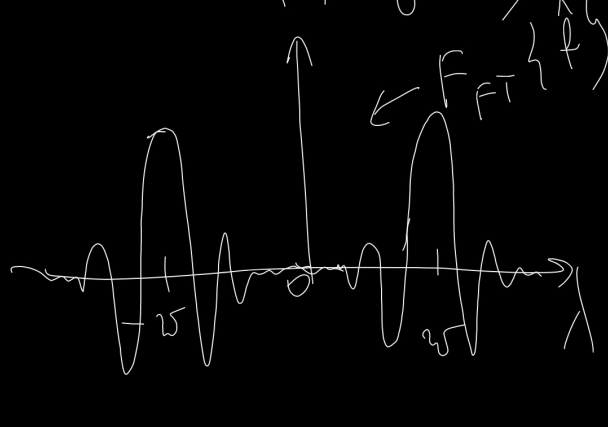
FT on finite time interval

$$\hat{f}(t) = f(t) \cdot g(t), \quad g(t) = \begin{cases} 1, & \text{if } a \le t \le b \\ 0, & \text{otherwise} \end{cases}$$

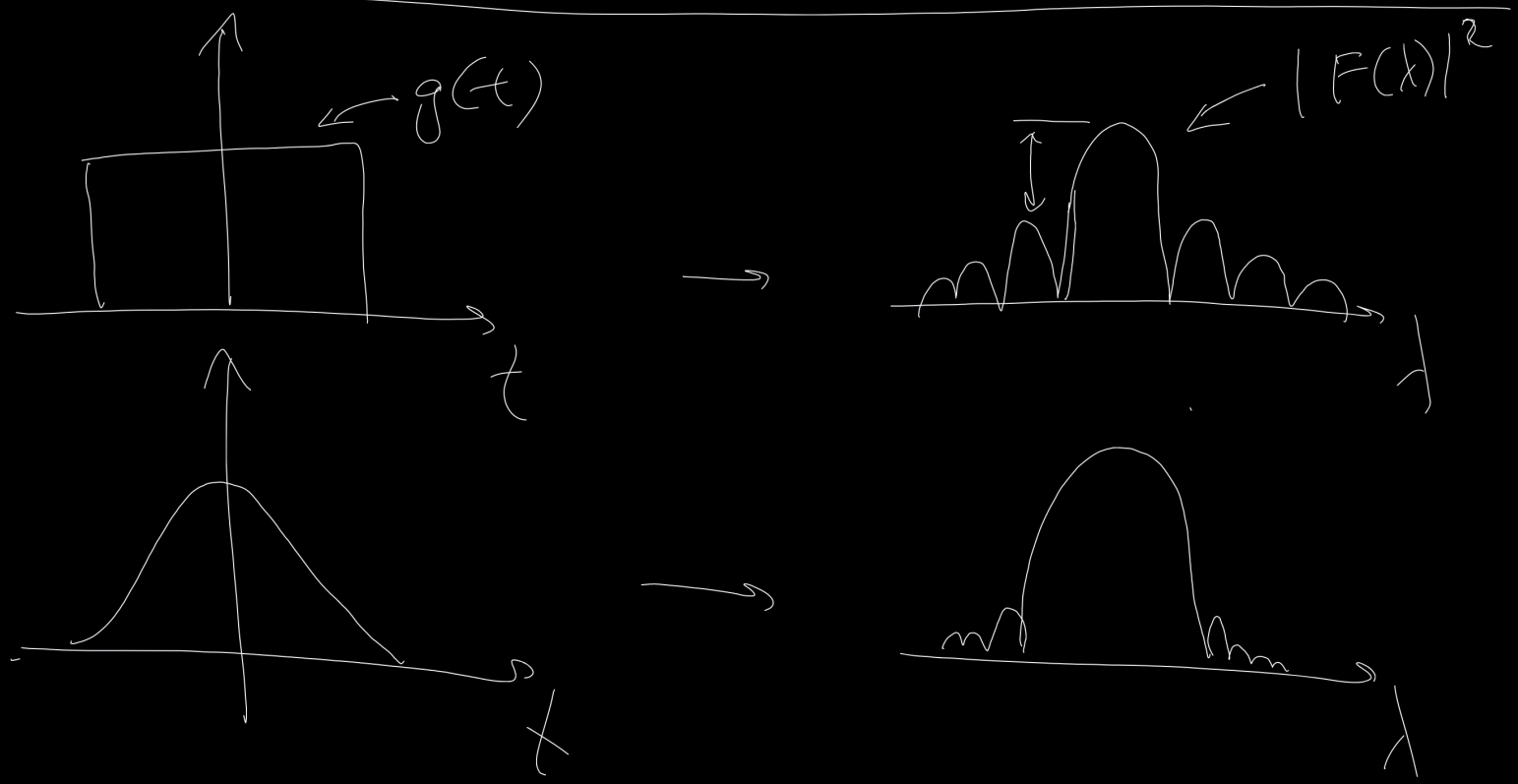
$$F_{FT}\{\hat{f}\}(\lambda) = F_{FT}\{f \cdot g\}(\lambda) = F_{FT}\{f\}(\lambda) * F_{FT}\{g\}(\lambda)$$







Ham window, Hamming window

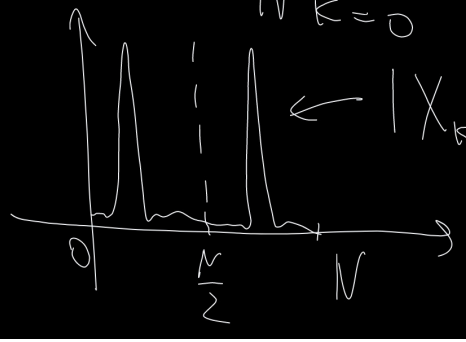


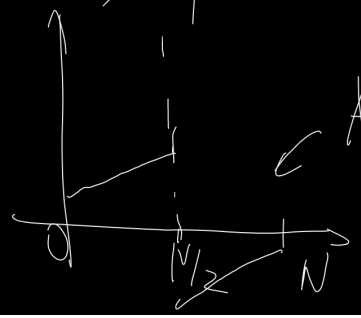
Discrete Fourier Transform (DFT)

$$x_0, x_1, \ldots, x_{N-1} \in \mathbb{R} \xrightarrow{DFT} X_0, X_1, \ldots, X_{N-1} \in \mathbb{C}$$

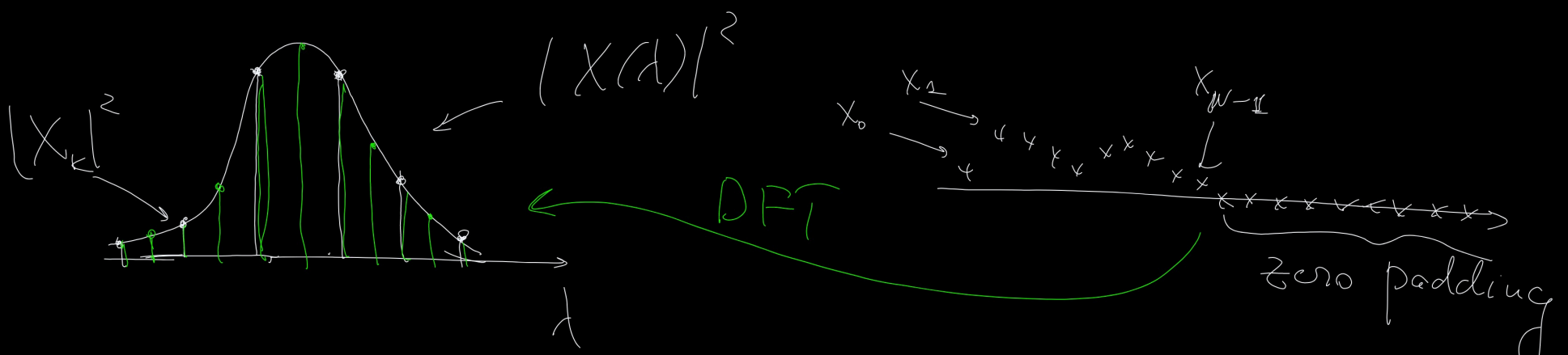
$$X_k = \sum_{n=0}^{N-1} x_n \exp\left(-i \underbrace{\frac{2\pi}{N} k \cdot n}_{=1}\right), \quad k = 0, \ldots, N-1$$

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k \exp\left(i \frac{2\pi}{N} k \cdot n\right), \quad n = 0, \ldots, N-1$$





$$X(\lambda) = \sum_{n=0}^{N-1} x_n \exp(-i\lambda n)$$

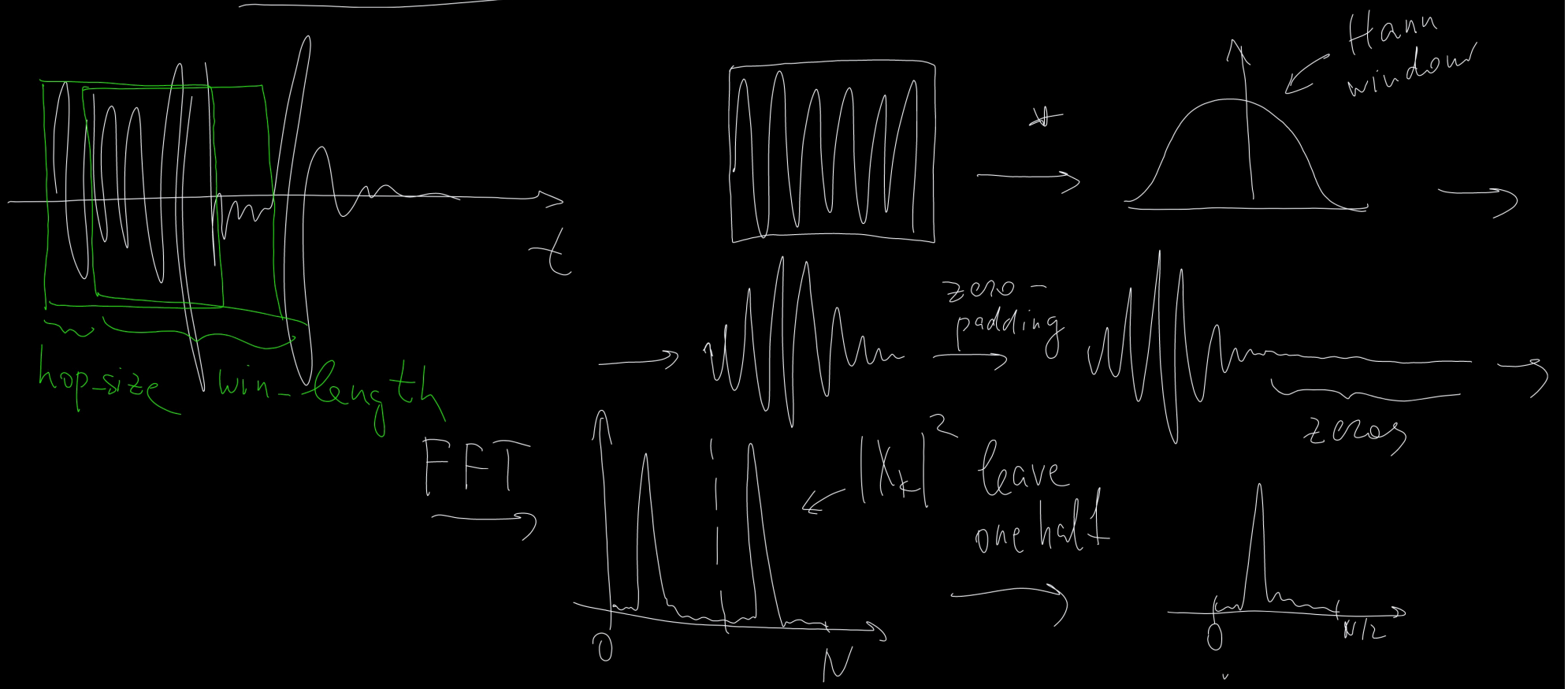


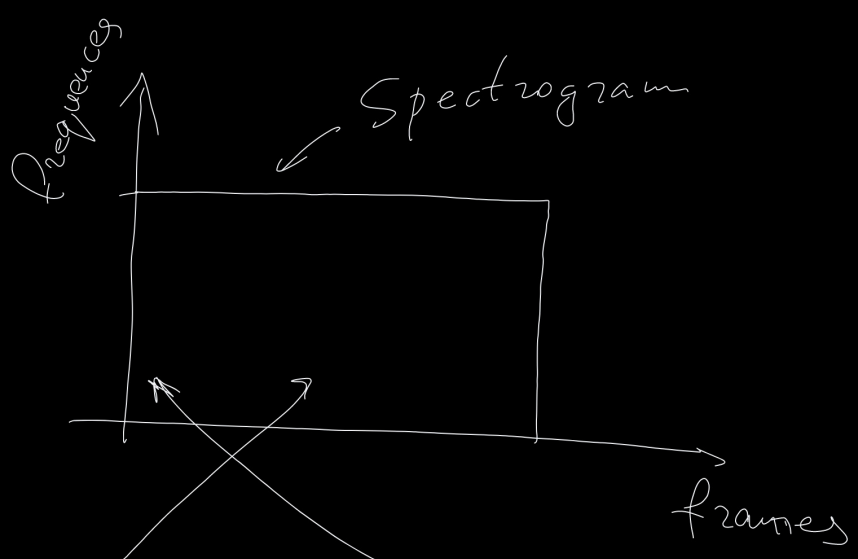
DFT computational complexity: $O(N^2)$

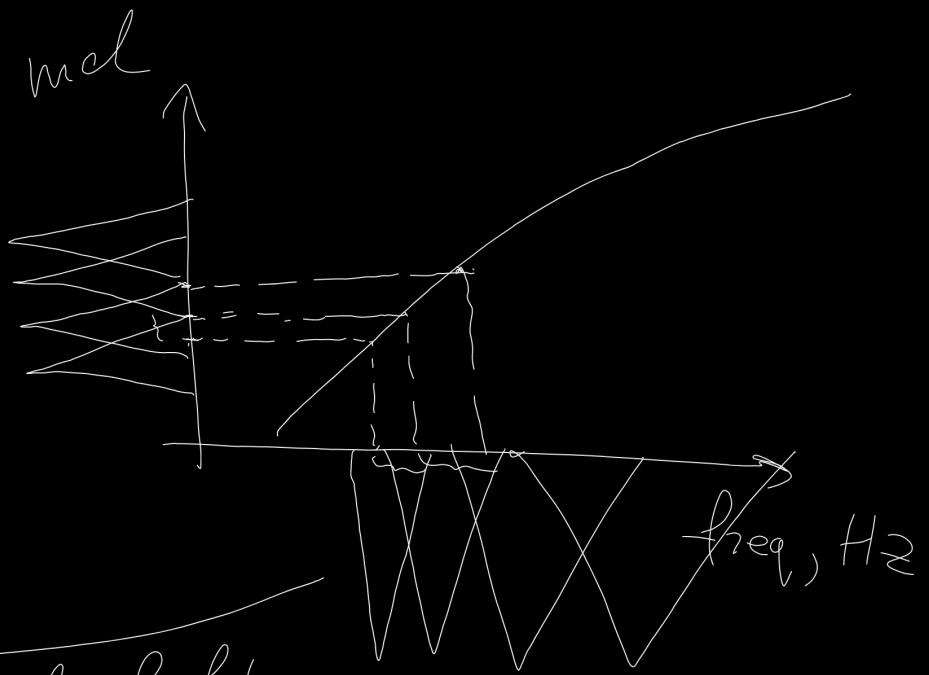
Fast Fourier Transform (FFT): $O(N \log N)$

$N = 2^m$

Audio Spectrogram







apply log
transform to amplitudes

bank of mel filters,
$n_mel = 40, 80$

log-mel-spectrogram

torchaudio.transforms.MelSpectrogram(data,

sample_rate=...,
win_length=...,
hop_size=...,
n_fft=...,
n_mels=...)

	Speech	Music
win_length	25 ms	93 ms
hop_size	10 ms	25 ms
n_fft	512, 1024	2048, 4096
n_mels	80	128